Serhii Liutov

Senior Data Scientist / AI Engineer

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SUMMARY

Senior Machine Learning Engineer and Data Scientist with **9+ years of experience** architecting **production-grade AI systems**, automated **ETL pipelines (dbt, SQL)**, and quantitative models across **Enterprise SaaS and Web3**. Specialized in **multi-agent LLM systems (RAG, LangGraph, Bedrock)**, recommendation engines, and scalable **MLOps** using **Python, FastAPI, Golang, and Docker**.

SKILLS SUMMARY

- **AI, LLMs & Agentic Systems:** LangGraph, LangChain, AWS Bedrock, OpenAI API, Google Gemini, Model Context Protocol (MCP), RAG, AI Agents, Prompt Engineering, Fine-Tuning, Hugging Face, RLHF.
- **Machine Learning & Data Science:** PyTorch, TensorFlow, Scikit-learn, XGBoost, Recommendation Systems, Anomaly Detection, Time Series (SARIMAX, GARCH), NLP (BERT), A/B Testing, SHAP (XAI).
- **Data Engineering & Cloud:** dbt, SQL (Advanced), Snowflake, Dune Analytics (Spellbook), PostgreSQL, GCP (BigQuery), AWS (S3, Lambda), Dagster, Trino, Power BI, Tableau, Airflow, ClickHouse.
- **Backend, MLOps & Infrastructure:** Python (Advanced), Golang, FastAPI, Docker, SQLAlchemy (Async), Pydantic v2, Alembic, CI/CD (GitHub Actions, GitLab CI), Streamlit, REST APIs.
- **Spoken Languages:** English (Advanced / C1), Ukrainian (Native), Russian (Native).

EXPERIENCE HIGHLIGHTS

UST Kyiv, Ukraine (Remote)

Senior Machine Learning Engineer (Consultant) *May 2025 – August 2026*

- AI Agent Architecture: Architected "StageCraft", an enterprise-grade multi-agent platform automating CI/CD migration (Jenkins to GitHub Actions/GitLab). Orchestrated **LangGraph** agents via **AWS Bedrock** for Context-Based Learning (CBL) and iterative code refactoring.

- Multi-Tenant Deployment: Migrated legacy architectures to Chainlit leveraging MCP (Model Context Protocol) and Playwright, delivering an adaptable AI agent system for multiple enterprise clients.

- Backend Engineering: Built a high-concurrency **FastAPI** microservice architecture using **Python** and **SQLAlchemy**. Solved critical race conditions via database constraints and optimized connection pooling for high-throughput agent workloads.

- Production Hardening: Enforced enterprise security standards including strict SSL/TLS database connections, environment-aware configuration, and sensitive credential redaction in telemetry logs.

- DevOps & Infrastructure: Standardized the development lifecycle using **Docker** and **Alembic**, automating schema migrations and integrating a full-stack local environment (Vite/React/Angular + FastAPI + Postgres).

- LLM Integration & RAG: Implemented RAG pipelines and HITL (Human-in-the-Loop) workflows to validate and refine AI-generated pipeline configurations.

Confidential Perpetual DEX Protocol Kyiv, Ukraine (Remote)

Freelance Data Scientist & BI Analyst *August 2024 – July 2026*

- Advanced Analytics: Built 30+ complex dashboards on **Dune Analytics** covering trader segmentation, user churn, stop-loss behavior, bridging costs, gas usage, and open interest analysis across multiple EVM chains.

- Financial Modeling: Designed SQL models tracking PnL vs Pool Value, borrow fee simulations, price impact metrics, and protocol fee projections.

- Derivative Performance Analysis: Conducted comprehensive analysis of derivative products (BTC, ETH, LP tokens) through fee tracking, utilization monitoring, liquidity analysis, and TVL trends.

- Competitive Benchmarking: Performed competitive benchmarking of other top perpetual DEXes, delivering actionable insights for feature prioritization, liquidity design, and user retention improvements.

- ETL Development: Maintained real-time hourly ETL pipelines using Dune's Spellbook architecture (Dune SQL, dbt, and Trino DB connector) to aggregate and normalize cross-chain data for protocol health monitoring.

Onchain Intelligence Engine Kyiv, Ukraine (Remote)
Senior Data Scientist (Project-Based) *September 2025*
- Recommendation System Engineering: Engineered a hybrid recommendation pipeline using **XGBoost** for opportunity ranking and **SHAP** for explainable AI, providing personalized DeFi yield strategies.
- Generative AI & LLM Integration: Developed an **LLM-powered** "Mood Explainer" service using **Google Gemini** and **LangSmith** to transform user behavioral data into human-readable narratives with strict JSON validation.
- High-Load API Architecture: Architected a scalable **FastAPI** microservice with **SQLAlchemy** and **PostgreSQL**, leveraging robust caching and database optimization to handle high-concurrency requests.
- Data Engineering: Implemented automated ETL jobs to normalize cross-chain DeFi yield data from sources like Coingecko and DefiLlama.
- DevOps: Fully containerized the application with **Docker** and managed database schema migrations via **Alembic** for reproducible deployments.

Thirdwave Labs Kyiv, Ukraine (Remote)
Senior Business Intelligence Data Scientist *October 2024 - May 2025*
- Data Pipeline Engineering: Built and maintained **Dagster** and **dbt** pipelines to collect and transform raw data into structured Snowflake analytics datasets.
- Analytics & KPIs: Calculated key behavioral indicators (Engagement Score, Retention Metrics) using Snowflake SQL to drive user segmentation and search optimizations.
- Visualization & Reporting: Created interactive **Power BI** dashboards to visualize platform distributions and leveraged AI features for automated narrative insights.
- Public Analytics: Replicated internal analytics pipelines for transparent public reporting and external stakeholders.

1kx Venture Kyiv, Ukraine (Remote)
Machine Learning Engineer (Project-Based) *September 2024 - November 2024*
- AI Platform Development: Engineered a Retrieval-Augmented Generation (**RAG**) AI platform using **LangChain** and **OpenAI** to query a multimodal knowledge base of 1,000+ proprietary projects.
- Autonomous Agents: Fine-tuned LLM-based agents for structured summarization, entity extraction, and automated metadata labeling.
- Multimodal Pipelines: Developed AI pipelines combining image-to-text models and PDF parsers for comprehensive due diligence.

Renormalizing Kyiv, Ukraine (Remote)
Machine Learning Engineer *June 2023– July 2024*
- **Generative AI & RAG:** Architected enterprise LLM chatbots utilizing OpenAI, LangChain, and Snowflake, implementing a Retrieval-Augmented Generation (RAG) framework for scalable corporate knowledge retrieval.
- **Applied NLP & LLM Fine-Tuning:** Fine-tuned LLMs and applied advanced prompt engineering (few-shot inference) to classify complex engineering activities. Developed NLP-driven algorithms to accurately match unstructured data across platforms (e.g., linking GitHub commits to Jira tasks).
- **Predictive Modeling & MLOps:** Engineered Monte-Carlo models to simulate engineering workflows, forecast team productivity, and calculate risk probabilities. Packaged and deployed these ML models as production-ready API endpoints.

AMLBot Kyiv, Ukraine (Remote)
Data Scientist *June 2022 – May 2023*
- On-Chain Forensics & Graph ML: Applied graph-based machine learning and network analysis across EVM blockchains to trace anomalous money flows and detect phishing activities, successfully identifying $12M in compromised funds.
- NLP & Entity Recognition: Developed Named Entity Recognition (NER) models using BERT to automatically extract entities and threat intelligence from unstructured documents.
- Quantitative Simulation: Built Monte-Carlo models to simulate trading activity on decentralized exchanges and estimate expected profit probabilities and risk.

MHP Kyiv, Ukraine (Remote)
Data Scientist *December 2019 – October 2022*
- **Time Series & Optimization:** Implemented ML anomaly detection on massive IoT climate sensor datasets. Solved complex linear optimization problems for facility climate control, deploying the solutions via interactive Streamlit web applications and Power BI.
- **Retail Analytics & Scoring:** Built logistic regression scoring models to predict retail outlet success and applied association rules (Market Basket Analysis) to optimize product matrices.

- **NLP & Predictive Modeling:** Deployed BERT-based NLP models to automatically classify large, unstructured product catalogs into specific taxonomic groups.
- **Data Engineering:** Engineered automated web-scraping pipelines to extract, aggregate, and monitor real-time commodity prices from multiple external sources.

CRYPTO BRIEFING Kyiv, Ukraine (Freelance/ Remote)
Data Analyst *December 2018 – April 2019*
- **Market Analysis:** Evaluated high-volume social and network data to perform qualitative and quantitative analysis of major technology ecosystems.

ATTIC LAB Kyiv, Ukraine
Data Analyst *March 2018 – November 2018*
- **Due Diligence Analytics:** Engineered data models to conduct comprehensive operational analysis, scam detection, and risk assessment of emerging tech startups.

KPMG Kyiv, Ukraine
Data Scientist/Risk Analyst *September 2017 – March 2018*
- **Risk Modeling:** Engineered Probability of Default (PD) Rate models in R and built credit scoring models utilizing logistic regression.
- **Data Engineering:** Orchestrated macroeconomic data pipelines, performed hypothesis testing, and built enterprise visualizations.

EDUCATION, TRAINING & CERTIFICATIONS

DATAROOT LABS
Data Science & AIMay 2019 – August 2019
Intensive program covering Applied Statistics, Deep Neural Networks (TensorFlow/Keras), and Advanced ML Algorithms.

KYIV SCHOOL OF ECONOMICS
Master of Arts in Economics (taught in English) *September 2016 – June 2018*

NATIONAL TECHNICAL UNIVERSITY OF UKRAINE "IGOR SIKORSKY KYIV POLYTECHNIC INSTITUTE"
Master of Arts in Management of Innovation *September 2016 – June 2018*

NATIONAL TECHNICAL UNIVERSITY OF UKRAINE "IGOR SIKORSKY KYIV POLYTECHNIC INSTITUTE"
Bachelor of Arts in Nuclear Engineering (with honors) *September 2012 – June 2016*

Coursera: Certificates in Machine Learning Algorithms.

INDEPENDENT PROJECTS

Quantitative Algorithmic Trading Engines | Python, AI/ML, Binance API, Telegram API
Independent Project | 2024 – Present
- **Automated Execution Pipelines:** Engineered multiple low-latency, systematic trading pipelines in Python that ingest automated webhook signals and translate them into deterministic order routing via direct exchange API integrations.
- **Algorithmic Risk Management:** Architected a programmatic risk-management framework enforcing strict capital preservation and progressive compounding, utilizing Machine Learning (LightGBM/Optuna) to filter signals based on statistically validated historical probabilities.
- **Operational Monitoring:** Designed an asynchronous monitoring layer utilizing the Telegram Bot API to dispatch real-time execution audits, state transitions, and risk exposure metrics, enabling continuous performance monitoring.